Employee-Department-Microservices

Project Description

This is a backend project developed using Java and Spring Boot. It manages employee and department details through REST APIs.

Technologies Used

- Java
- Spring Boot
- Spring Web
- Spring Data JPA
- Hibernate
- MySQL
- H2 Database
- Maven
- Postman
- Spring Tool Suite (STS)

Modules

1. Employee Module

Used to manage employee details.

- Create employee
- Get employee details
- Get all employees
- Update employee details
- Delete employee

Technologies used: Java, Spring Boot, REST API, JPA, Hibernate, and MySQL/H2.

2. Department Module

Used to manage department details.

- Create department
- Get department details
- Get all departments
- Update department details
- Delete department

Technologies used: Java, Spring Boot, REST API, JPA, Hibernate, and MySQL/H2.

3. Exception Handling

Used to handle errors such as employee or department records not being found and invalid input.

Database

We have configured two databases:

- **MySQL:** Used to store data permanently.
- **H2:** An in-memory database used for testing and development.

The active database is selected through the Spring profile configuration.

Project Structure

```
com.company
├── CompanyApplication.java
├── employee
│   ├── Employee.java
│   ├── EmployeeController.java
│   ├── EmployeeService.java
│   └── EmployeeRepository.java
├── department
│   ├── Department.java
│   ├── DepartmentController.java
│   ├── DepartmentService.java
```

```
│   └── DepartmentRepository.java
└── exception
```

How to Run

1. Import the project into Spring Tool Suite.
2. Configure the database connection.
3. Update the Maven project.
4. Run `CompanyApplication.java`.
5. Test the REST APIs using Postman.

Conclusion

This project helps us understand Spring Boot REST APIs, database connectivity, JPA, Hibernate, and the separation of application logic into Employee and Department modules.